

NED University of Engineering and Technology
Department of Computer Science and Information Technology
Self-Practice Questions

Q.1 Determine whether the following function is injective, surjective, or a bijection from $\mathbb{R}$ to $\mathbb{R}$. Justify your answer.

1. $f(x) = (3x - 2)/(x + 4)$
2. $f(x) = (2x + 7)/(x - 3)$
3. $f(x) = (2x - 3)/(x + 1)$

Q.2 Prove both distributive laws of set theory for the following sets:

- $A = \{x | x \text{ is a multiple of 5 and less than 30}\}$
- $B = \{x | x \text{ is an even number less than 12}\}$
- $C = \{x | x \text{ is a prime number less than 15}\}$

Q.3 Translate the following statement into propositional logic using the propositions provided. Then, for each condition, state its converse, contrapositive, and inverse: *"I will go to university only if I wake up early and the bus arrives on time, and if it rains, I will stay at home or study online."*

(Use: U: Go to university, E: Wake up early, B: Bus arrives on time, R: It rains, H: Stay at home, S: Study online)

Q.4 Prove De Morgan's Laws for Quantifiers for the following statements:

- a) *There is a student in this class who studies daily.*
- b) *Every student in this class studies daily.*

Q.5. a. Are these system specifications consistent? *"The system is in multiuser state if and only if it is operating normally. If the system is operating normally, the kernel is functioning. The kernel is not functioning or the system is in interrupt mode. If the system is not in multiuser state, then it is in interrupt mode. The system is not in interrupt mode."*

b. Are these system specifications consistent? *"The router can send packets to the edge system only if it supports the new address space. For the router to support the new address space it is necessary that the latest software release be installed. The router can send packets to the edge system if the latest software release is installed, the router does not support the new address space."*

Q.6. a. Show that $(p \vee q) \rightarrow (p \vee q) \equiv T$ by developing a series of logical equivalences.

b. Show that $(p \vee (p \wedge q)) \equiv p$ using a series of logical equivalences.

Q.7 Write an algorithm for **bubble sort** and sort $[z, x, y, a, b, c]$ showing the list obtained at each step. Mention the time complexity.